RESEARCH & DEVELOPMENT · ENVIRONMENTAL ENGINEERING

# ARUNA

## Kinetic-based design of an IoT-enabled household anaerobic digester for food waste

Anaerobic Recycling and Upcycling of Nutrient-rich food wAste: sized from mass balance and Gompertz kinetics, instrumented with NDIR methane sensing and safety interlocks.

SDG 7.1 SDG 11.6 SDG 12.3 SDG 13.2

**[Team name]** · [Supervisor] · [Institution] · 2026



# Food waste is Indonesia's largest organic waste stream.

**53 kg**
household food waste per capita per year
UNEP (2024)

**14.7 Mt**
household food waste per year, Indonesia
UNEP (2024)

**≈40%**
of municipal solid waste is food waste
SIPSN KLHK (2023)

**1.7 Gt**
$CO_2e$ from food loss & waste, 2000–2019
Bappenas (2021)

## Why anaerobic digestion?

| Property of food waste | Typical value | Source |
|---|---|---|
| Total solids (TS) | 23–27 % | Xu et al. (2018) |
| Volatile solids (VS/TS) | 90–95 % | Xu et al. (2018) |
| Biochemical $CH_4$ potential | 330–490 mL $CH_4$/g VS | Zhang et al. (2007) |
| Theoretical (Buswell) | ≈ 643 mL $CH_4$/g VS | Buswell & Mueller (1952) |

## Research questions

- **RQ1** What digester volume, OLR and HRT keep household food-waste digestion stable?
- **RQ2** How much biogas, cooking time and avoided $CO_2e$ can one unit deliver?
- **RQ3** Can sensor feedback make the process usable for non-experts (SUS ≥ 68)?

**Sources:** UNEP Food Waste Index Report 2024; SIPSN KLHK 2023; Bappenas 2021 FLW study; Zhang et al. 2007, Bioresour. Technol. 98:929; Xu et al. 2018, Bioresour. Technol. 247:1047.

# Current options trade energy, space or visibility.

| Approach | Footprint / scale | Input | Energy balance | Monitoring | Main limitation |
|---|---|---|---|---|---|
| Electric food recycler (Lomi) | countertop | food scraps | consumes 0.6–1.0 kWh/cycle | app | no energy recovery |
| HomeBiogas 2 | 2.1 × 1.15 m, 2.0 $m^3$ | 4 L FW + 4 L water/d | up to 2 h cooking/d | visual | outdoor, 2.4 $m^2$ footprint |
| Fixed-dome digester (BIRU) | masonry, outdoor | livestock manure | household cooking gas | none | needs manure; 25,157 units (2021) |
| IoT digester research | pilot / lab | mixed organics | measured, not productised | ESP32: T, pH, $CH_4$, $H_2S$ | not a household device |
| **ARUNA** | **0.44 × 0.44 m, 100 L** | **1.5 kg FW + 1.5 L water/d** | **≈43 min cooking/d (predicted)** | **NDIR $CH_4$, pH, T, level + interlocks** | **to be field-validated** |

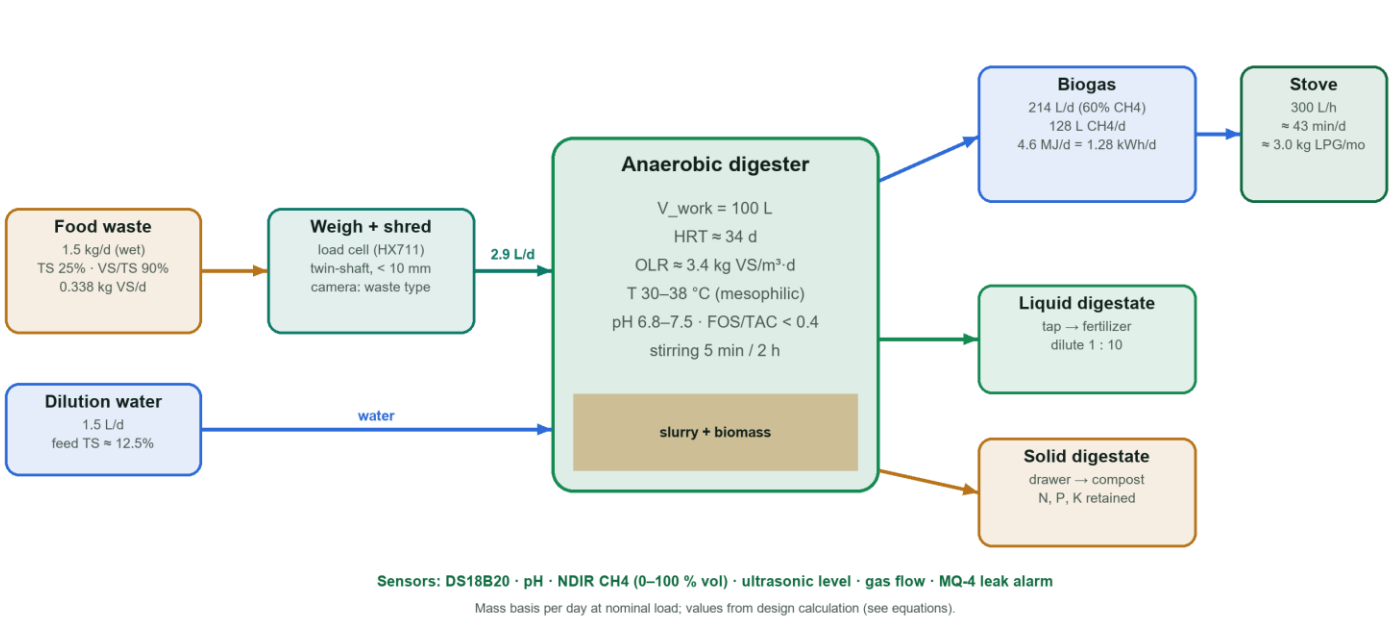
## Research gap

No household option combines food-waste input, energy recovery, indoor footprint and real-time process monitoring. Low-cost MQ-type gas sensors (300–10,000 ppm) cannot quantify biogas $CH_4$ (50–70 % vol); an NDIR sensor (0–100 % vol) is required.

**Sources:** Lomi (2024) energy use; HomeBiogas (2024) HB2 specifications; ESDM/Hivos BIRU (2021); Prasetyono et al. 2026, J. Mechatron. Electr. Power Veh. Technol. 17(1):29; Rajendran et al. 2012, Energies 5:2911; Winsen MQ-4 & MH-440D datasheets.

04 · SOLUTION

# Shred, stir, digest, measure: three outputs.




**H1** At OLR ≤ 3.4 kg VS/m³·d and HRT ≈ 34 d: $CH_4$ ≥ 50 %, ≥ 0.30 m³ $CH_4$/kg VS, FOS/TAC < 0.4. **H2** SUS ≥ 68.

1. **Biogas**
   214 L/d · 1.3 kWh/d
2. **Liquid digestate**
   N-rich fertilizer, dilute 1:10
3. **Solid digestate**
   compost drawer

**SDG 7.1**
clean cooking fuel ≈ 3.0 kg LPG-eq/month

**SDG 11.6**
0.55 t/yr diverted from municipal waste

**SDG 12.3**
food waste valorised at source

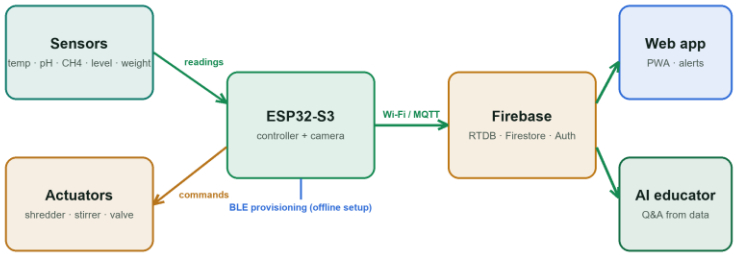
**SDG 13.2**
0.72 t $CO_2$e/yr avoided (potential)

**Sources:** Design calculation (this work) using Zhang et al. 2007; Xu et al. 2018; IPCC 2006 Vol. 5 Ch. 3; IPCC AR5 GWP100 = 28; UN A/RES/70/1 (SDG targets).

# A 0.19 m² digester with a digital twin.



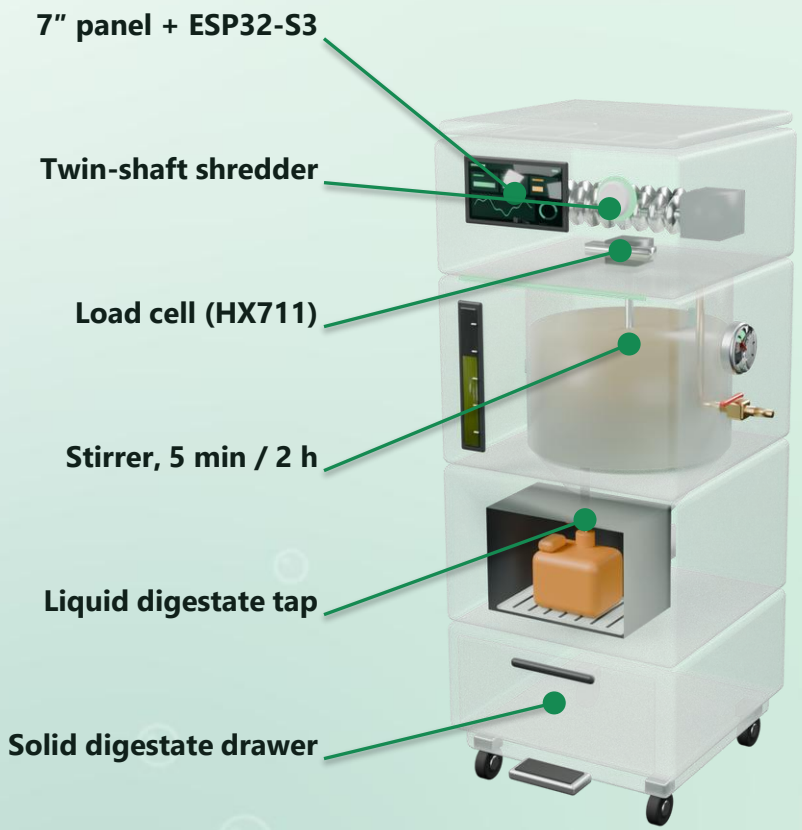

| Hardware | Value |
|---|---|
| Footprint × height | 44 × 44 × 112 cm |
| Working volume | 100 L |
| Nominal / peak load | 1.5 / 3.0 kg FW/d |
| HRT / OLR | 34 d / 3.4 kg VS/$m^3$·d |
| Controller | ESP32-S3 (Wi-Fi, BLE) |
| Process sensors | T, pH, NDIR $CH_4$, level, mass |
| Safety | MQ-4 leak alarm, valve |




| Software | Implementation |
|---|---|
| Telemetry | 15-min sampling → Firebase RTDB |
| Interface | PWA, live at aruna-id.web.app |
| Guidance | interlock alerts + AI educator |
| Setup | BLE provisioning, offline |

**Sources:** This work (3D model & live app); Espressif ESP32-S3 datasheet; Google Firebase Realtime Database documentation.

# Gravity-fed layout, instrumented for process control.

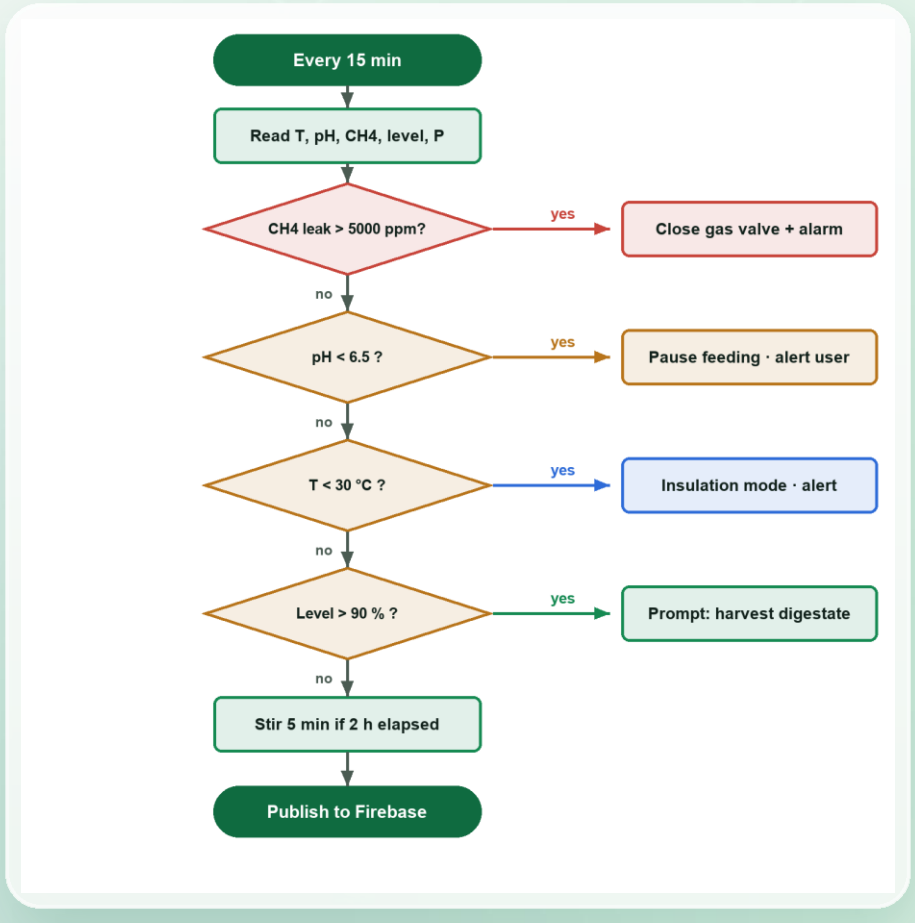



| Parameter | Sensor | Range | Accuracy / role |
|---|---|---|---|
| Temperature | DS18B20 | −55–125 °C | ±0.5 °C |
| $CH_4$ in biogas | NDIR MH-440D | 0–100 % vol | T90 < 30 s |
| $CH_4$ leak | MQ-4 | 300–10,000 ppm | alarm 10 % LEL |
| pH | glass probe | 0–14 | 3-pt calibration |
| Level | ultrasonic | 0–100 % fill | ≈ ±1 cm |
| Input mass | load cell + HX711 | 0–5 kg | 24-bit ADC |
| Gas volume | diaphragm meter | L/d | reference |

**Safety & process interlocks**
leak > 5,000 ppm → close valve · pH < 6.5 → pause feeding · T < 30 °C → insulation alert · level > 90 % → harvest prompt

**Sources:** Maxim/Analog Devices DS18B20 datasheet; Winsen MH-440D (NDIR, 0–100 % vol) & MQ-4 (300–10,000 ppm) datasheets; $CH_4$ LEL 5 % vol.

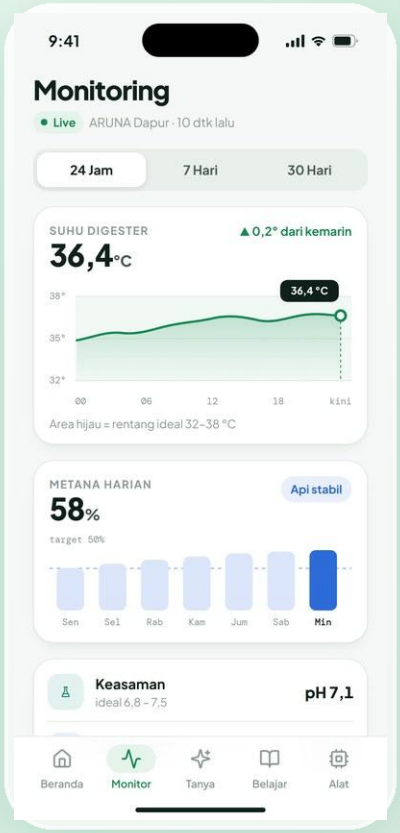
# From raw signals to plain-language decisions.







Status




Trends + ideal band




AI educator

**Sources:** This work (rule-based controller, live PWA). Rule thresholds: pH 6.5–7.5 and FOS/TAC < 0.4 (Chen et al. 2008, Bioresour. Technol. 99:4044; Liu et al. 2021, Appl. Sci. 11:11843).

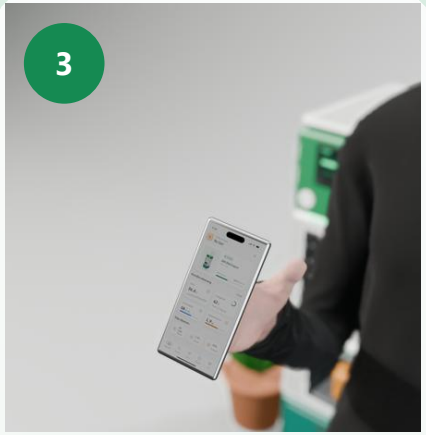
# One day of operation, quantified.



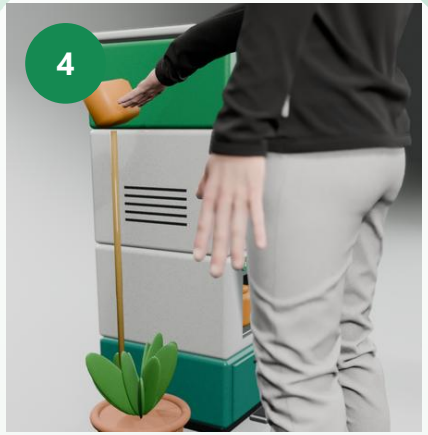

**Feed**
1.5 kg FW + 1.5 L water; weighed & shredded




**Digest**
OLR 3.4 kg VS/m³·d, stir 5 min / 2 h




**Monitor**
15-min telemetry; alert if pH < 6.5



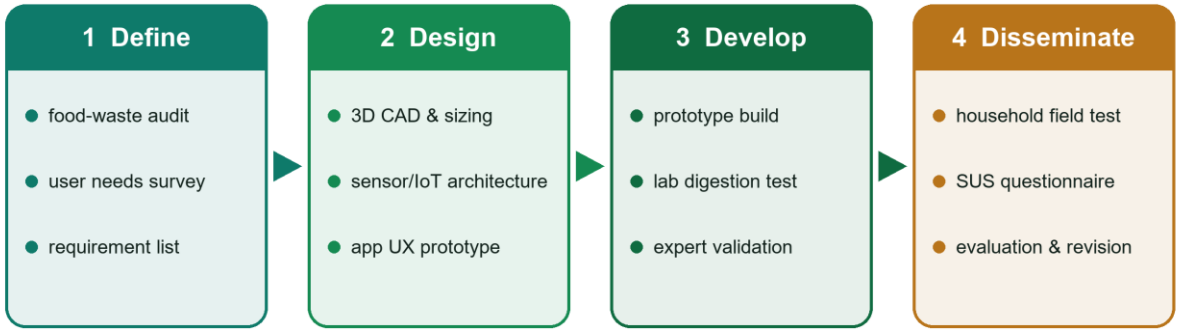

**Fertilize**
liquid digestate diluted 1:10




**Cook**
214 L biogas ≈ 43 min at 0.30 m³/h

**Sources:** Design calculation (this work); stove demand 0.2–0.45 m³/h (SSWM/Akvopedia, Biogas combustion, 2019).

# R&D (4D model) with quantitative exit criteria.




**Mass balance**

$$OLR = F \cdot TS \cdot (VS/TS) / V_w \qquad HRT = V_w / Q$$

**Kinetics (Gompertz)**

$$M(t) = P \cdot \exp\{-\exp[(R_m \cdot e/P)(\lambda - t) + 1]\}$$

**Theoretical $CH_4$ (Buswell)**

$$C_cH_hO_oN_n + H_2O \rightarrow CH_4 + CO_2 + NH_3$$

**Energy & cooking**

$$E = V_{CH4} \cdot 35.8\ MJ/m^3\ ;\ t_{cook} = V_{biogas} / q_{stove}$$

**Avoided landfill $CH_4$ (IPCC)**

$$W \cdot DOC \cdot DOC_f \cdot MCF \cdot F \cdot 16/12 \cdot (1 - OX)$$

| Experiment | Design | Standard / method | Output |
|---|---|---|---|
| BMP assay | ISR ≥ 2, 37 °C, triplicate | VDI 4630 | P, Rm, λ |
| Semi-continuous | OLR 2.0 / 3.4 / 4.5 × 3 reps, 2 HRT | APHA 2540 G; FOS/TAC | yield, stability |
| Field test | [n] households × 30 d | telemetry + logbook | $CH_4$ %, L/d, kg/d |
| Usability | post-test questionnaire | SUS (Brooke 1996) | score 0–100 |

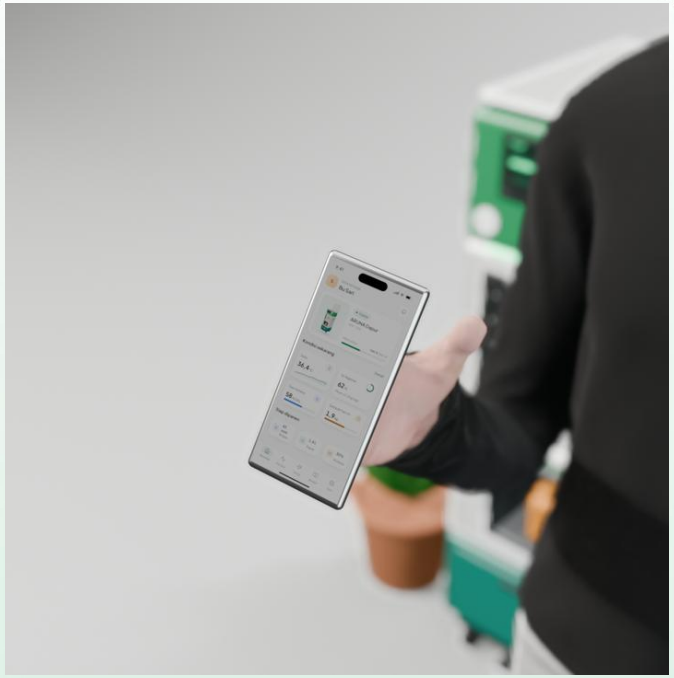
- **Kinetics:** non-linear least squares, $R^2$, RMSE
- **Comparison:** one-way ANOVA + Tukey, α = 0.05
- **Biodegradability:** BMP(exp) / BMP(theoretical)

**Sources:** Thiagarajan et al. 1974; Buswell & Mueller 1952, Ind. Eng. Chem. 44:550; Zwietering et al. 1990, Appl. Environ. Microbiol. 56:1875; VDI 4630 (2016); APHA 2540 (2017); Brooke 1996.

# Measurement plan with acceptance criteria.

ILLUSTRATIVE DATA · replace with field results

| Parameter | Instrument / method | Frequency | Acceptance | Basis |
|---|---|---|---|---|
| Temperature | DS18B20 | 15 min | 30–38 °C | Weiland (2010) |
| pH | calibrated probe | 15 min | 6.8–7.5 | Chen et al. (2008) |
| FOS/TAC | two-point titration | weekly | < 0.4 | Liu et al. (2021) |
| $CH_4$ content | NDIR + [ref. analyser] | continuous | ≥ 50 % vol | Xu et al. (2018) |
| Biogas volume | diaphragm gas meter | daily | ≥ 171 L/d | this work (−20 %) |
| TS / VS | APHA 2540 G | weekly | VS removal ≥ 70 % | Mata-Alvarez (2000) |
| Digestate N, P, K | lab analysis | end of cycle | [standard] | national fertilizer standard |
| Usability | SUS | end of test | ≥ 68 | Bangor et al. (2009) |



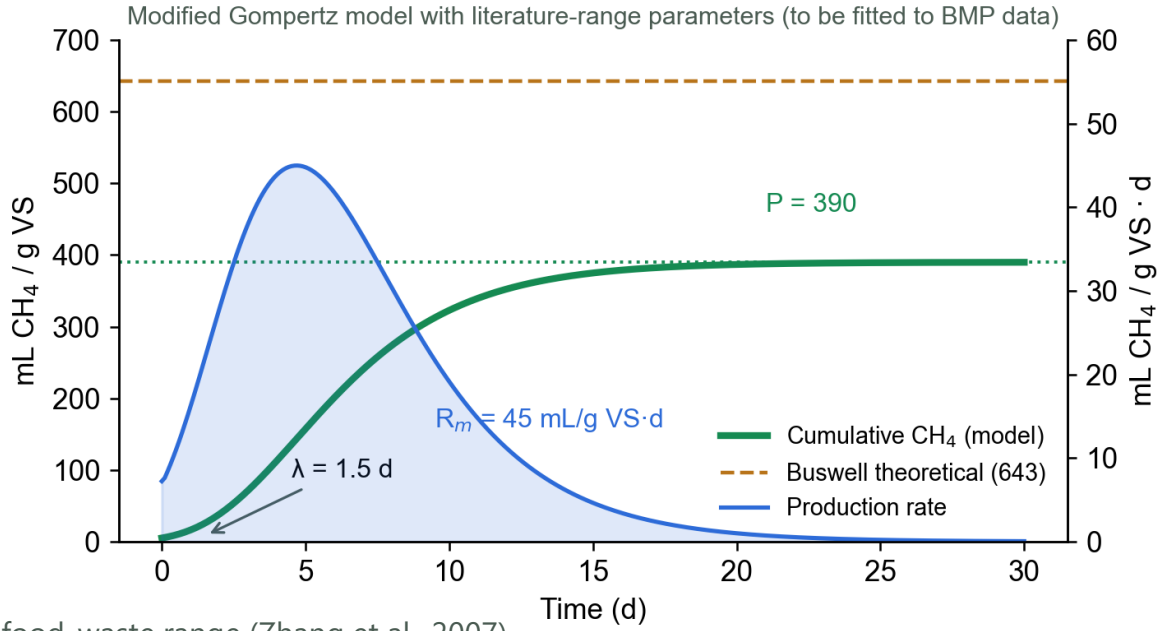
Rendered scenario; replace with field photos.

**Sources:** Weiland 2010; Chen et al. 2008; Liu et al. 2021, Appl. Sci. 11(24):11843; Xu et al. 2018; Mata-Alvarez et al. 2000, Bioresour. Technol. 74:3; APHA 2540 (2017); Bangor et al. 2009, J. Usability Stud. 4:114.

# Design predictions against the literature.

PREDICTED FROM DESIGN CALCULATION · measured data pending

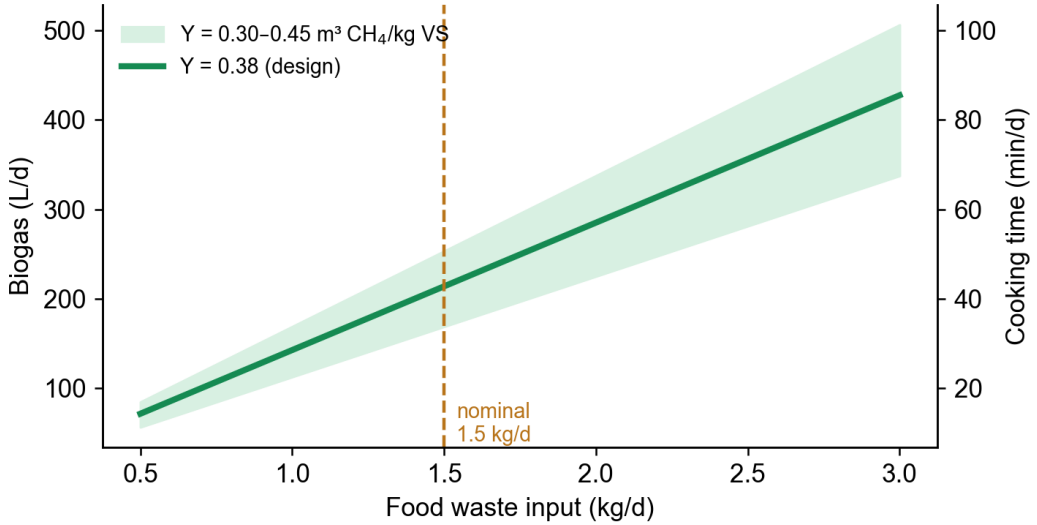
| Parameter | Household of 4 | Nominal | Peak |
|---|---|---|---|
| Input (kg FW/d) | 0.58 | **1.5** | 3.0 |
| OLR (kg VS/m³·d) | 1.3 | **3.4** | 6.8 |
| HRT (d) | 88 | **34** | 17 |
| Biogas (L/d) | 83 | **214** | 428 |
| Cooking (min/d) | 17 | **43** | 86 |
| LPG-eq (kg/month) | 1.2 | **3.0** | 6.0 |
| Avoided (t $CO_2$e/yr) | 0.28 | **0.72** | 1.44 |




- **Biodegradability ≈ 61 %:** BMP 390 vs Buswell 643 mL $CH_4$/g VS, within reported food-waste range (Zhang et al., 2007).
- **Stability:** nominal OLR 3.4 is within stable ranges; peak OLR 6.8 at HRT 17 d risks VFA accumulation → pH interlock (Chen et al., 2008).
- **Scale:** a household of 4 yields ≈ 17 min/d; nominal load fits extended families or small eateries.

**Sources:** Design calculation (this work); Gompertz parameters from literature ranges, to be fitted (Zwietering et al. 1990); Zhang et al. 2007; Chen et al. 2008; UNEP 2024 (53 kg/cap/yr).

# From prototype to verified impact.




**1** **0–6 months · Validate**
- BMP + semi-continuous tests; fit P, Rm, λ
- calibrate NDIR vs gas analyser

**2** **6–12 months · Optimise**
- co-digestion to C/N 20–30
- $H_2S$ scrubber; insulation for T < 25 °C

**3** **12–36 months · Scale**
- on-device AI waste recognition
- MBG school-kitchen pilots with MRV

**Sources:** Sensitivity analysis (this work); C/N 20–30 and $H_2S$ removal (Weiland 2010, Appl. Microbiol. Biotechnol. 85:849; Mata-Alvarez et al. 2000).

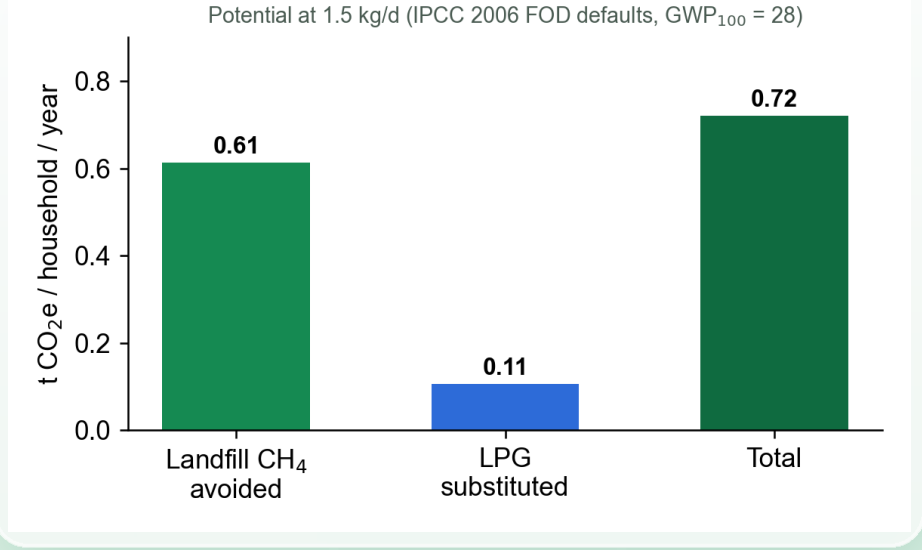
# Answers to the research questions.

**RQ1** A 100 L digester at OLR 3.4 kg VS/m³·d and HRT ≈ 34 d is predicted to be stable for 1.5 kg/d; peaks above ≈ 3 kg/d require pausing.

**RQ2** ≈ 214 L biogas/d → ≈ 43 min cooking, ≈ 3.0 kg LPG-eq/month and ≈ 0.72 t $CO_2e$/yr avoided.

**RQ3** NDIR sensing, rule-based interlocks and app guidance close the monitoring gap; SUS to be measured.

**Live prototype app** **aruna-id.web.app**
Values are design predictions; measured results will replace them after testing.




**Sources:** This work; IPCC 2006 Vol. 5 Ch. 3 (FOD defaults); IPCC AR5 (GWP100 = 28).

# Literature and data sources.


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